

Magnetic Force

Building Models to Describe our World

Magnetic Force, Fields and Torques



Outline

The Magnetic Force

- Magnetism Basics

- Recall: vector-cross product

- Right hand rule(s)

- Force on a moving charge

- Magnetic force on a charged particle

- Right hand rule for vector/cross product

The Magnetic Field

- Electron in a magnetic field

- Force on a current-carrying wire

- Example: force from a wire

The Magnetic Torque

- Torque on a current-carrying loop

- Magnetic moment

- The Hall effect

- The sign of charge carriers

- The Lorentz force and velocity selector

- The mass spectrometer

- Example: mass spectrometer at airports

- Force on a current-carrying wire (again)

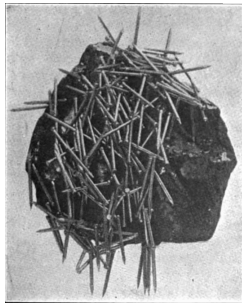
- Force on a curved wire

The Magnetic Force

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Magnetism

- Some rocks were found to be “magnetic” (rather, they were found to attract and repel each other, and they came from a place called Magnesia); “lodestone”.
- If you make a needle out of a piece of magnetic material, and suspend it, it kind of points towards the geographic North Pole of the Earth.
- We call a piece of magnetic material a “magnet”. We label the side that points towards Earth’s North Pole, the North side of the magnet (cause it points North).



<https://en.wikipedia.org/wiki/Lodestone>



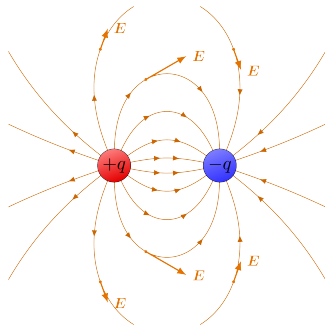
<https://www.walmart.ca/en/ip/Pocket>

Magnets

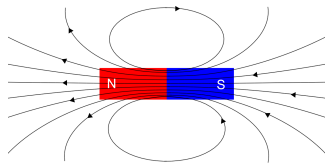
- With bar magnets, one side (=N) of the magnet always wants to point towards geographic north (ish).
- We call the other side of the magnet S, since it points to geographic South (ish).
- It has been observed that the same sides of magnets repel each other (NN or SS), while the opposite sides attract (N-S).

Magnetic fields

- Fields were introduced as a tool to model “force at a distance”.
- Magnetic **field points in the direction of force on N** (from N to S).
- Just like electric field lines:
 - Field vector length represented by density of lines.
 - Field vector is tangent to the lines.
- Unlike electric field lines:
 - **No isolated magnetic charges**, only bar magnets (dipoles).
 - **Field lines never end**, they are always closed loops.
- The closest analogy with electric charges is that bar magnets are like electric dipoles.

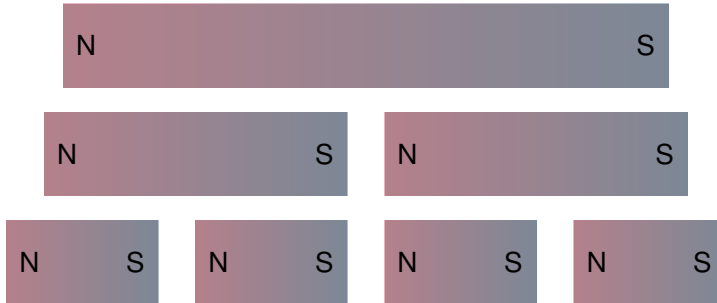


An electric dipole.



A magnetic dipole.

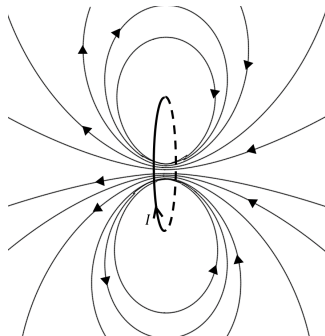
Cutting up a bar magnet



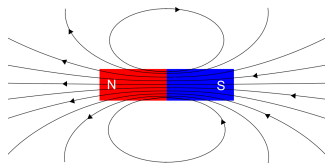
When we cut a magnet in 2, we get another magnet (with N and S) sides.

Tiny bar magnets

- You can think of magnetic materials as being made up of tiny little bar magnets.
- The fields for those “tiny bar magnets” add up to make the field from a large bar magnet.
- Those tiny little bar magnets ultimately come from electrons spinning around in atoms:
 - As we will see in next chapter, moving charges create magnetic fields.
 - The electrons spin around the atom.
 - The electrons also spin about their own axis (this is actually the main contribution, and it's Quantum Mechanics).



Magnetic field from a loop of current.



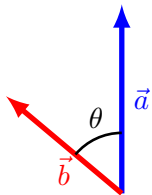
Magnetic field from a bar magnet.

Recall: vector cross-product

- If we have two vectors, $\vec{a}$ and $\vec{b}$, we can define the cross-product:

$$\vec{c} = \vec{a} \times \vec{b}$$

- $\vec{c}$ is perpendicular to both $\vec{a}$ and $\vec{b}$.
 - and has magnitude: $c = ab \sin \theta$
- You can always move vectors around, so easiest to draw them tail to tail before taking cross product
- Use right hand rule to figure which way $\vec{c}$ points

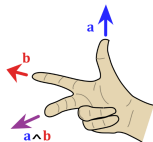


Vectors, $\vec{a}$ and $\vec{b}$.

Two right-hand rules

1. A cross-product between two vectors.
Right-hand rule for cross-product.

$$\vec{c} = \vec{a} \times \vec{b}$$



Right-hand rule for cross-product.

2. When you need to relate rotation to a vector.
Right-hand rule for axial vectors.



Right-hand rule for axial vectors.

Force on a moving charge

- If a moving charge is in a magnetic field, it experiences a force:

$$\vec{F} = q\vec{v} \times \vec{B}$$

- The magnitude of the force is:

$$F = qvB \sin \theta$$

- We ultimately use this to define the magnetic field (which is clearly not as pretty as for the electric field). That is, we infer B from the magnitude of the force, F .

Work done by a constant magnetic field on a moving charge

- The force is always perpendicular to the velocity:

$$\vec{F} = q\vec{v} \times \vec{B}$$

- The displacement is always parallel to the velocity,
 - The force is thus always perpendicular to the displacement (scalar product is 0)
- The magnetic force can do no work; it cannot change the kinetic energy.
- The magnetic force can only change the direction of the velocity, not its magnitude.

Magnetic force on a charged particle

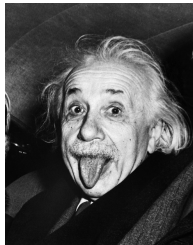
$$\vec{F} = q\vec{v} \times \vec{B}$$

- Some observations:
 - Always perpendicular to both $\vec{v}$ and $\vec{B}$ (cross-product).
 - Can do no work.
 - Proportional to electric charge.
 - No magnetic force on a charge at rest.

Magnetic force on a charged particle

$$\vec{F} = q\vec{v} \times \vec{B}$$

- Some observations:
 - Always perpendicular to both $\vec{v}$ and $\vec{B}$ (cross-product).
 - Can do no work.
 - Proportional to electric charge.
 - No magnetic force on a charge at rest. → This should make you uncomfortable, why?



https://en.wikipedia.org/wiki/Albert_Einstein

FYI Magnetic force on a charged particle

- How does the magnetic force know that humans are predominantly right-handed? For example, how would you explain the right-hand rule to an Alien, who has no concept of right and left?

FYI Magnetic force on a charged particle

- Remember: There is no such thing as a force, a magnetic field, an electric field, etc. They are all just mathematical constructs.

FYI Magnetic force on a charged particle

- Physical Laws cannot depend on humans being right-handed:
 - We observe that when an electron is near a source of magnetic field (e.g. a current carrying wire), it moves in a certain direction.
 - We use our right hand to define a thing called the magnetic field.
 - We use our right hand again to determine the force from the magnetic field.
- Any physical observation (e.g. which way an electron is deflected) depends on using your right hand twice → Then, you could also use your left hand, as once you do two cross products, it doesn't matter which hand you used (don't try this at home, unless you are 100% sure that you understand the argument!)

Force on a moving charge

- If a moving charge is in a magnetic field, it experiences a force:

$$\vec{F} = q\vec{v} \times \vec{B}$$

- The magnetic force:
 - Only exerted on a moving charge.
 - Does no work.
 - Perpendicular to magnetic field.
 - Perpendicular to velocity.
 - Magnitude: $qvB \sin \theta$.
 - Direction: RHR.

Right hand rule for vector/cross product

The cross-product between two vectors.

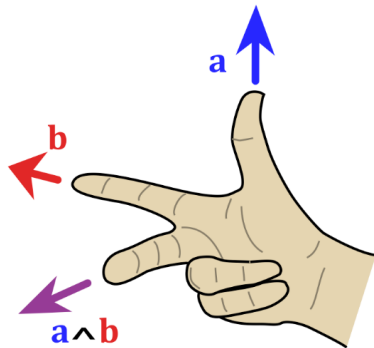
$$\vec{c} = \vec{a} \times \vec{b}$$

Magnitude: $ab \sin \theta$

Direction: Right hand rule. The product is always perpendicular to the plane formed by $\vec{a}$ and $\vec{b}$.

It does not commute:

$$\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$$



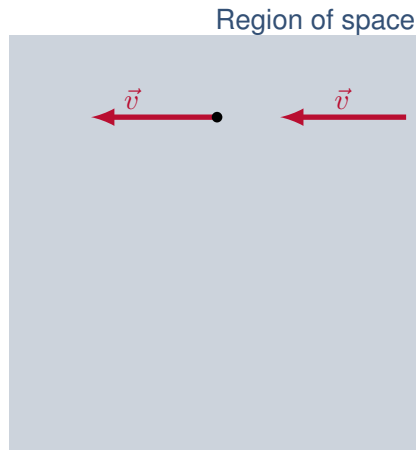
Right-hand rule for cross-product.

The Magnetic Field

Building Models to Describe our World
Magnetic Force, Fields and Torques

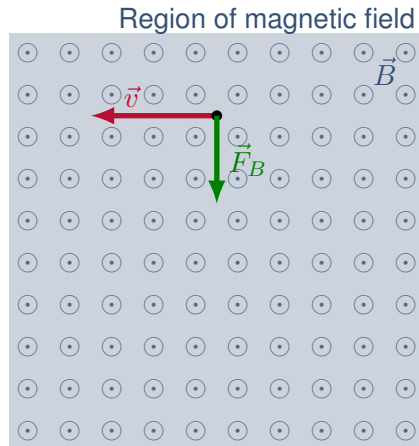
Electron in a magnetic field

- An electron enters a region of space



Electron in a magnetic field

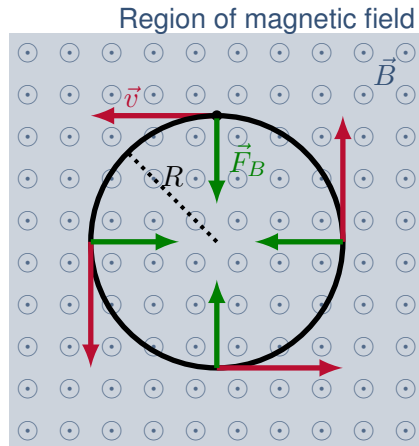
- An electron enters a region of space with a magnetic field perpendicular to its velocity
- The magnetic force is always perpendicular to the velocity:



Electron in a magnetic field

- An electron enters a region of space with a magnetic field perpendicular to its velocity
 - The magnetic force is always perpendicular to the velocity:
 - Speed is constant (no work).
 - Force is constant in magnitude and perpendicular to velocity.
- Uniform circular motion:

$$qvB = m \frac{v^2}{R}$$
$$\therefore R = \frac{m}{q} \frac{v}{B}$$

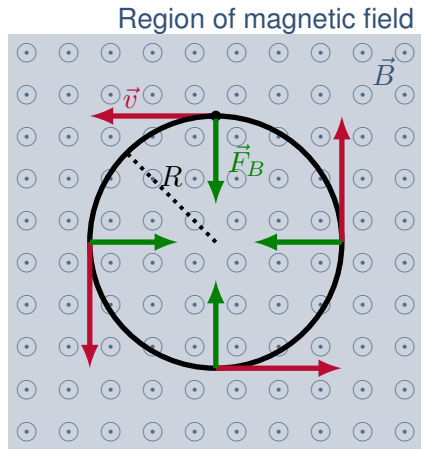


Electron in a magnetic field

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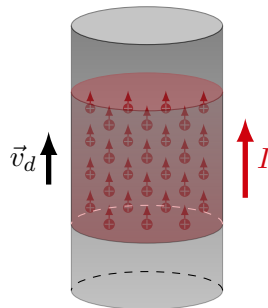
$$qvB = m \frac{v^2}{R}$$
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If the electron has a component of velocity in/out of the page, the path will be a helix.



Force on a current-carrying wire

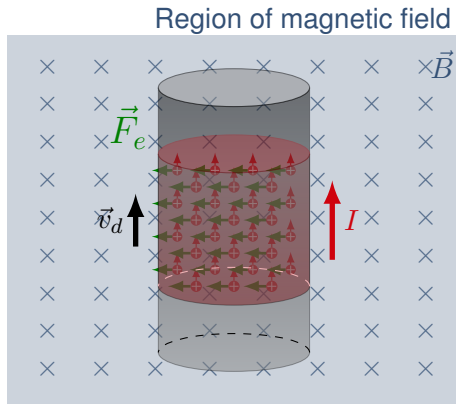
- Current: positive charges moving with a drift velocity, $\vec{v}_d$.



Positive charges drifting with an average velocity, $\vec{v}_d$. Would you come to different conclusions modelling electrons drifting downwards?

Force on a current-carrying wire

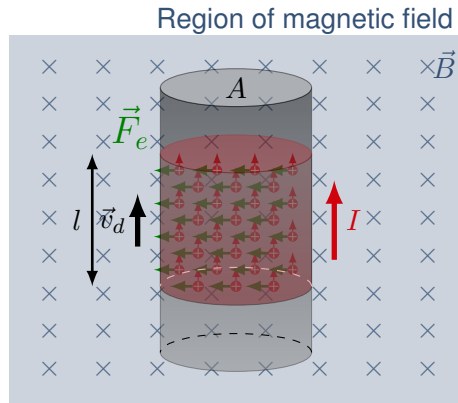
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$$\vec{F}_e = e\vec{v}_d \times \vec{B}$$



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Force on a current-carrying wire

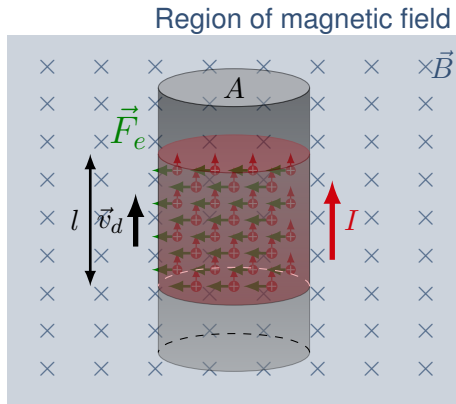
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- In a segment of length l , and area A , if there are n charges per unit volume, there are a total of N charges: $N = nAl$



Positive charges drifting with an average velocity, $\vec{v}_d$. Would you come to different conclusions modelling electrons drifting downwards?

Force on a current-carrying wire

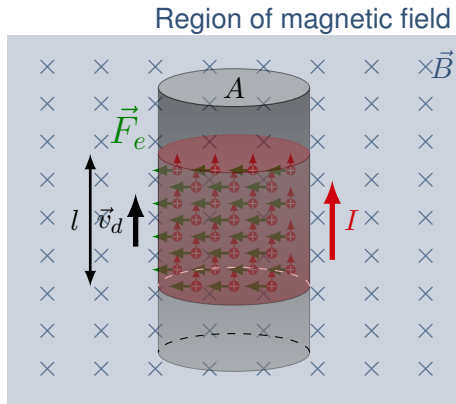
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- In a segment of length l , and area A , if there are n charges per unit volume, there are a total of N charges: $N = nAl$
- The total force on those charges is:
$$\vec{F} = N\vec{F}_e = nAle\vec{v}_d \times \vec{B}$$



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- Recall, microscopic model of current,
$$I = nAev_d$$

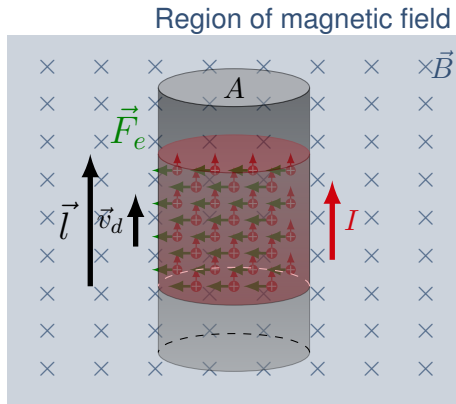


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 $\vec{F} = N\vec{F}_e = nAle\vec{v}_d \times \vec{B}$
- Recall, microscopic model of current,
 $I = nAev_d$
- Make the vector, $\vec{l}$, in the same direction as, $\vec{v}_d$:

$$\boxed{\vec{F} = I\vec{l} \times \vec{B}}$$



Positive charges drifting with an average velocity, $\vec{v}_d$. Would you come to different conclusions modelling electrons drifting downwards?

Force on a current-carrying wire (in general)

- On a section of length, dl :

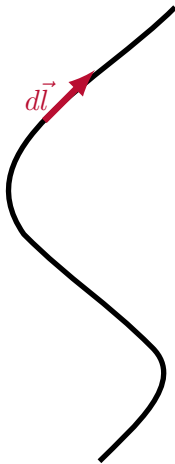
$$d\vec{F} = I d\vec{l} \times \vec{B}$$

- On a wire made of many sections:

$$\vec{F} = \int d\vec{F} = I \int d\vec{l} \times \vec{B}$$

- If it's just a straight wire of length, l , and the field is uniform

$$\vec{F} = I \int d\vec{l} \times \vec{B} = I \vec{l} \times \vec{B}$$

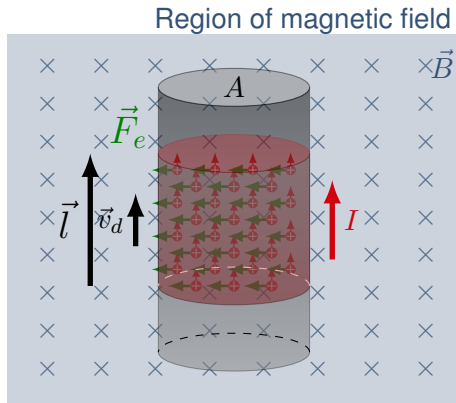


The vector $d\vec{l}$ points in the same direction as the current.

Wire of length, l , carrying current, I , in a magnetic field, $\vec{B}$.

- Current: positive charges moving with a drift velocity, $\vec{v}_d$.
- One positive charge e feels a force:
 $\vec{F}_e = e\vec{v}_d \times \vec{B}$
- In a segment of length l , and area A , if there are n charges per unit volume, there are a total of N charges: $N = nAl$
- The total force on those charges is:
 $\vec{F} = N\vec{F}_e = nAle\vec{v}_d \times \vec{B}$
- Recall, microscopic model of current,
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- Make the vector, $\vec{l}$, in the same direction as, $\vec{v}_d$:

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Force on a current-carrying wire (in general)

- On a section of length, dl :

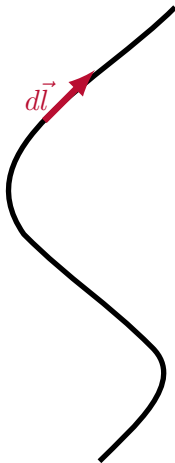
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The Magnetic Torque

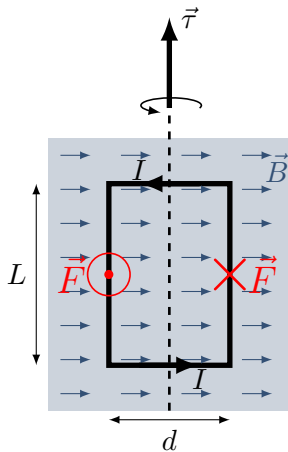
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Magnetic Force, Fields and Torques

Torque on a current-carrying loop

- The force on either segment:
 $F = ILB$
- The torque from one segment:
 $\tau_s = \left(\frac{d}{2}\right) F = \frac{d}{2} ILB$
- The total torque (when there is no magnetic flux through the loop):
 $\tau = 2\tau_s = IdLB = IAB = \mu B$
- We introduce the **magnetic moment**, μ , of a current carrying loop of area A :

$$\mu = IA$$
$$\therefore \tau = \mu B$$

The torque must depend on the orientation of the loop...



A rectangular loop carrying counter-clockwise current, I .

Magnetic moment

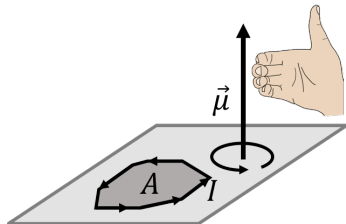
- Define the magnetic moment, $\vec{\mu}$, for a loop of area A carrying current I :

$$\vec{\mu} = I\vec{A}$$

- Direction of $\vec{A}$, $\vec{\mu}$, given by normal to the surface, direction of current, and right hand rule for axial vectors.
- Torque on a loop given by:

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

- This is analogous to the torque on an electric dipole from an electric field (electric dipole moment $\vec{p} \rightarrow \vec{\mu}$ magnetic moment)

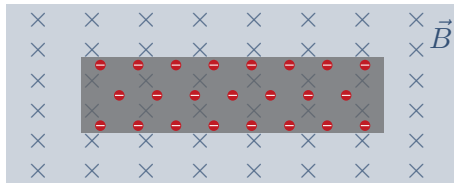


Using the right-hand rule for axial current to determine the direction of the magnetic moment of a loop carrying current, I .

The net torque on the loop always acts to make $\vec{\mu}$ and $\vec{B}$ parallel.

The Hall Effect

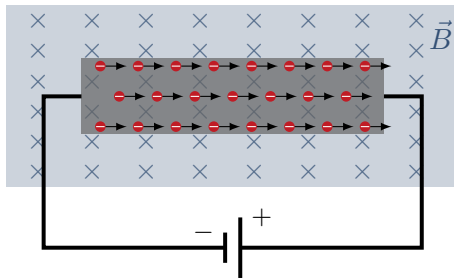
- We have a strip of metal immersed in a magnetic field into the page, as shown.
 1. The electrons are at rest in the metal that is immersed in the magnetic field.



A piece of conducting material in a magnetic field.

The Hall Effect

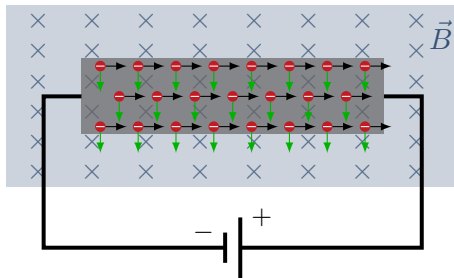
- We have a strip of metal immersed in a magnetic field into the page, as shown.
 1. The electrons are at rest in the metal that is immersed in the magnetic field.
 2. Apply a potential difference to the metal, electrons start moving.



A piece of conducting material in a magnetic field. A current is driven by a potential difference

The Hall Effect

- We have a strip of metal immersed in a magnetic field into the page, as shown.
 1. The electrons are at rest in the metal that is immersed in the magnetic field.
 2. Apply a potential difference to the metal, electrons start moving.
 3. Electrons are deflected downwards because of the magnetic field.

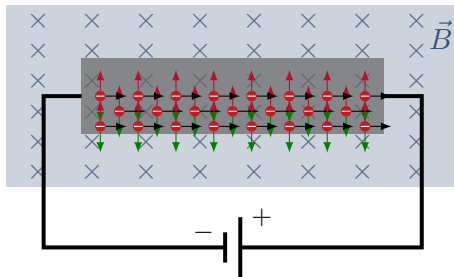


A piece of conducting material in a magnetic field. A current is driven by a potential difference. Negative charges are deflected downwards.

The Hall Effect

- We have a strip of metal immersed in a magnetic field into the page, as shown.
 1. The electrons are at rest in the metal that is immersed in the magnetic field.
 2. Apply a potential difference to the metal, electrons start moving.
 3. Electrons are deflected downwards because of the magnetic field.
 4. The bottom of the strip becomes more negative resulting in an **electric force** opposite of the magnetic force.

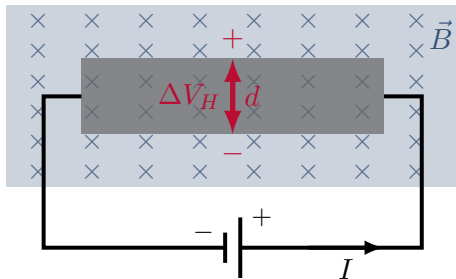
$$\vec{F}_E + \vec{F}_B = 0$$



A piece of conducting material in a magnetic field. A current is driven by a potential difference. Negative charges are deflected downwards. This causes a buildup of negative charge on the bottom.

The Hall Effect

- We have a strip of metal immersed in a magnetic field into the page, as shown.
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 2. Apply a potential difference to the metal, electrons start moving.
 3. Electrons are deflected downwards because of the magnetic field.
 4. The bottom of the strip becomes more negative resulting in an **electric force** opposite of the magnetic force.
 5. This leads to the Hall voltage, ΔV_H .



A piece of conducting material in a magnetic field. A current is driven by a potential difference. Negative charges are deflected downwards. This causes a buildup of negative charge on the bottom, leading a potential difference between the top and bottom, the "Hall Voltage".

The Hall Effect

By measuring the Hall voltage, you can measure B . A Hall probe!

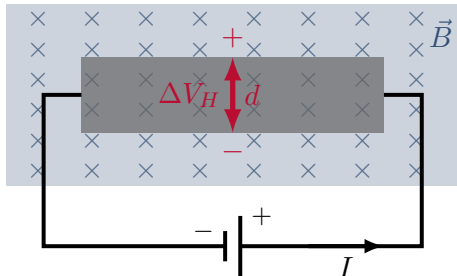
$$\vec{F}_E + \vec{F}_B = 0$$

$$\therefore qE = qvB$$

$$\frac{\Delta V_H}{d} = vB = \frac{I}{neA}B$$

$$\therefore \Delta V_H = \frac{Id}{neA}B$$

where v is the drift velocity and we made use of our model for current.

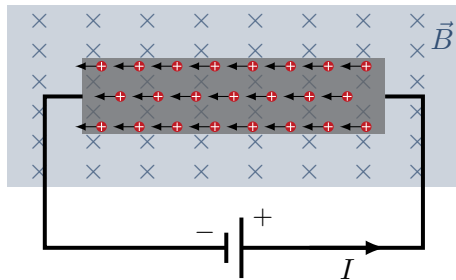


A piece of conducting material in a magnetic field. A current is driven by a potential difference. Negative charges are deflected downwards. This causes a buildup of negative charge on the bottom, leading a potential difference between the top and bottom, the “Hall Voltage”.

The sign of charge carriers

Can we tell if current is carried by positive or negative charges?

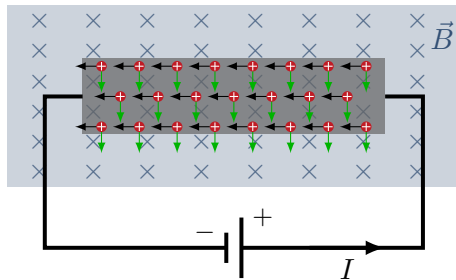
- What happens if it's actually positive charges carrying current? Their speed is in the opposite direction.



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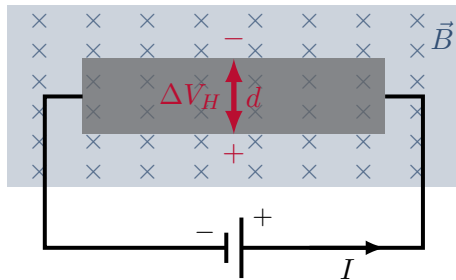
- What happens if it's actually positive charges carrying current? Their speed is in the opposite direction.
- The magnetic force is downwards, leading to a net positive charge at the bottom of the strip.



The sign of charge carriers

Can we tell if current is carried by positive or negative charges?

- What happens if it's actually positive charges carrying current? Their speed is in the opposite direction.
- The magnetic force is downwards, leading to a net positive charge at the bottom of the strip.
- The Hall voltage is now the opposite sign! By measuring the Hall voltage, you can determine that it is negative charges that are actually moving!



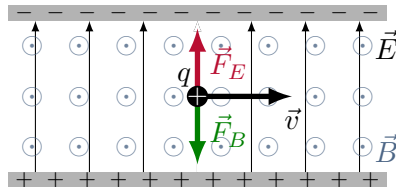
The Lorentz force and velocity selector

- If a charge is placed in both magnetic and electric fields, it will feel a force (the Lorentz Force):

$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$$

- We can build a “velocity selector” by having perpendicular electric and magnetic fields.
- Only particles with the “selected velocity” make it through without hitting the plates:

$$qE = qvB$$
$$\therefore v = \frac{E}{B}$$



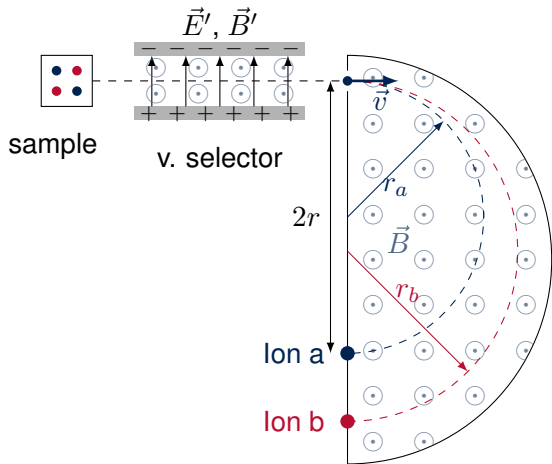
A velocity selector is designed with perpendicular electric and magnetic fields. The resulting forces cancel if q has the right velocity.

The mass spectrometer

- Material is vaporized, ionizing to be accelerated towards the velocity selector. Only those with speed, v , enter the mass spectrometer:

$$qvB = m \frac{v^2}{r}$$
$$\therefore \frac{m}{q} = \frac{rB}{v} = \frac{rBB'}{E'}$$

- Since $q = \pm ne$ for ions, one can determine the mass of the ions in the sample, hence the name.

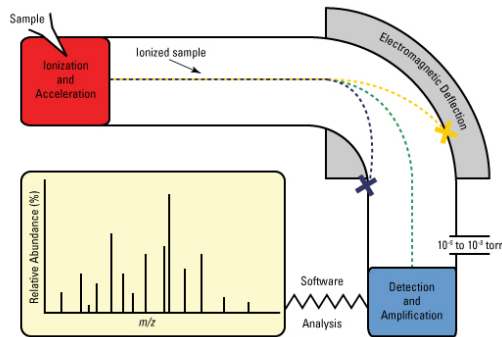


A mass spectrometer can determine the isotopic abundance of a sample by measuring the radius of ion trajectories in a magnetic field.

Example: Mass spectrometer at airports



https://commons.wikimedia.org/wiki/File:Okęcie_-_urządzenie_01.JPG



In practice, can count ions only at a certain position and vary the magnetic field.

Again: force on a current-carrying wire (in general)

- On a section of length, dl :

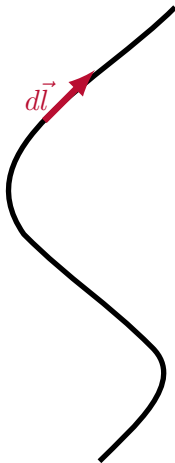
$$d\vec{F} = I d\vec{l} \times \vec{B}$$

- On a wire made of many sections:

$$\vec{F} = \int d\vec{F} = I \int d\vec{l} \times \vec{B}$$

- If it's just a straight wire of length, l , and the field is uniform

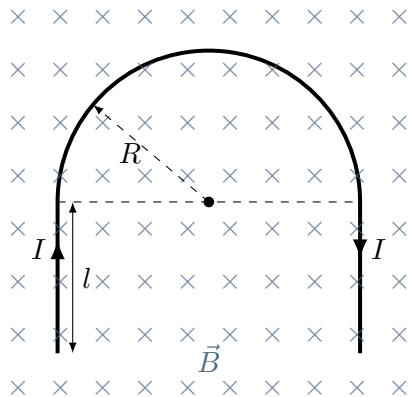
$$\vec{F} = I \int d\vec{l} \times \vec{B} = I \vec{l} \times \vec{B}$$



The vector $d\vec{l}$ points in the same direction as the current.

Force on a curved wire

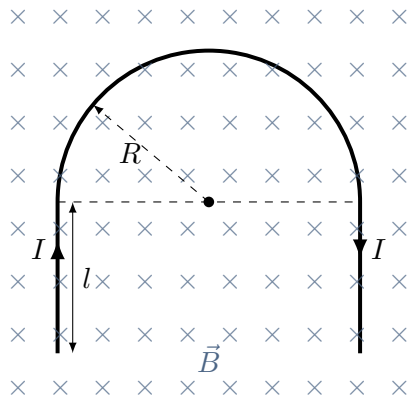
- Want to know total force on the wire.



A curved current-carrying wire in a magnetic field.

Force on a curved wire

- Want to know total force on the wire.
- The forces from the straight sections cancel.

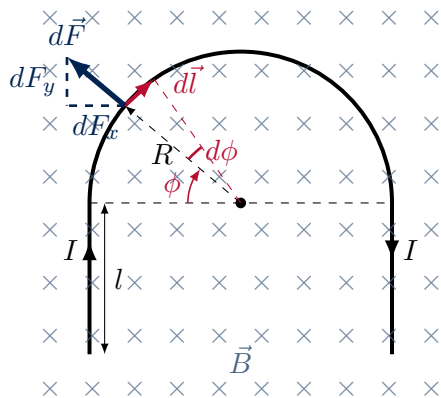


A curved current-carrying wire in a magnetic field.

Force on a curved wire

- Want to know total force on the wire.
- The forces from the straight sections cancel.
- Choose a segment, of length $dl = R d\phi$, it will experience a force:

$$d\vec{F} = I d\vec{l} \times \vec{B} = IBR d\phi (-\cos \phi \hat{x} + \sin \phi \hat{y})$$



A curved current-carrying wire in a magnetic field.

Force on a curved wire

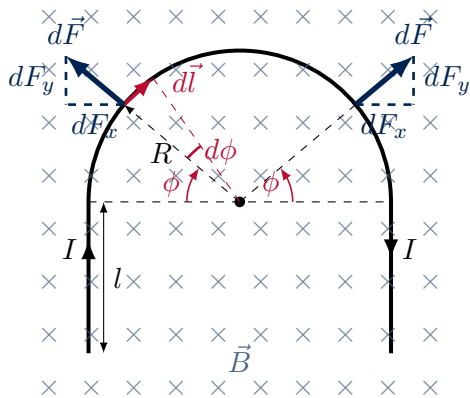
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- The x components of force cancel by symmetry. The total force in y :

$$F_y = \int_0^\pi IBR \sin \phi d\phi = IBR$$

You can easily convince yourself that the net force on a closed circular loop is zero!



A curved current-carrying wire in a magnetic field.